

The Ultimate Python Setup & Conda Guide

Welcome! This comprehensive guide will cover installation for all operating systems, environment management with Conda, and seamless integration with VS Code and Jupyter Notebooks.

1. 📁 Choosing Your Distribution: Anaconda vs. Miniconda

Before installing, you must decide which version of Conda you need.

Feature	Miniconda (Recommended)	Anaconda
Size	Small (~100 MB)	Massive (3 GB+)
Philosophy	Minimalist - Install only what you need.	"Batteries Included" - Everything pre-installed.
Speed	Fast to download and install.	Slower due to size.
Control	Keeps your system clean and lean.	Can feel cluttered with unused tools.

[!TIP] **Start with Miniconda.** It keeps your workspace organized. You can always install any package later using `conda install <package_name>`.

2. 🛠️ Installation Guide (All OS)

🍏 macOS & 🐧 Linux

1. **Download the installer:**
 - **macOS:** Download the [Apple Silicon \(M1/M2/M3\)](#) or [Intel](#) `.pkg` file.
 - **Linux:** Download the `.sh` script from the [Miniconda site](#).

2. **Run the installation:**
 - On macOS, just double-click the `.pkg`.
 - On Linux (or macOS terminal), run:

```
bash Miniconda3-latest-Linux-x86_64.sh
```

3. **Initialize Conda:** During installation, when asked *"Do you wish the installer to initialize Miniconda3 by running conda init?"*, type `yes` and press Enter.

4. **Restart Terminal:** Close and reopen your terminal to see `(base)` at the prompt.

Windows

1. **Download:** Get the [Windows 64-bit installer](#).

2. **Run Installer:**

- **Crucial Step:** On the "Advanced Options" page, **DO NOT** check "Add Anaconda to my PATH".
- **Check** "Register Anaconda as my default Python".

3. **Link to PowerShell:**

- Open **Anaconda Prompt** from the Start Menu.
- Run: `conda init powershell`

4. **Fix Security Policy:**

- Right-click Start -> **PowerShell (Admin)**.
- Run: `Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser`
- Type `Y` and Enter.

5. **Restart:** Close and reopen PowerShell. You should see `(base)` .

3. 🐍 Conda Tutorial: Environment Mastery

Never work in the `(base)` environment. Create a dedicated space for every project.

⚡ Essential Commands

Action	Command
Create Env	<code>conda create --name myenv python=3.10 -y</code>
Activate	<code>conda activate myenv</code>
Deactivate	<code>conda deactivate</code>
List Envs	<code>conda env list</code>
Install Package	<code>conda install pandas numpy</code>
Remove Env	<code>conda env remove --name myenv</code>

Best Practice: Reproducibility

To share your project with others, export your environment:

```
conda env export > environment.yml
```

Others can recreate it using:

```
conda env create -f environment.yml
```

4. VS Code & Jupyter Integration

To use your Conda environments inside VS Code notebooks:

1. Install VS Code Extensions:

- Open VS Code, go to Extensions (`Ctrl+Shift+X`).
- Install **Python** and **Jupyter**.

2. Prepare Your Environment: Activate your env and install the kernel bridge:

```
conda activate myenv  
conda install ipykernel -y
```

3. Select the Kernel in VS Code:

- Open a `.ipynb` file.
- Click "**Select Kernel**" (top right).
- Choose **Python Environments** -> **myenv**.

5. Testing & Verification

Once setup, run these tests to ensure everything is perfect.

✓ Terminal Check

Run `conda --version` . If it prints a version number, you're good!

✓ Notebook Check

Create a new cell in your Jupyter Notebook and run:

```
import sys  
import pandas as pd  
import numpy as np  
  
print(f"Python Path: {sys.executable}") # Should point to your 'myenv' folder  
print(f"Pandas: {pd.__version__}")  
print(f"NumPy: {np.__version__}")
```

6. Bonus: Lightweight Virtual Envs (`venv`)

If you don't want to use Conda for a small project, Python has a built-in tool:

- **Create:** `python -m venv .venv`
- **Activate (Mac/Linux):** `source .venv/bin/activate`
- **Activate (Windows):** `.venv\Scripts\activate`